Specifications for the Description of Barley Germplasm Resources

Author-prepared English reference translation

This reference translation follows the Chinese crop descriptor specification used for the CropMaestro study. It is not an official English edition issued by the source platform. Original section numbers and category-code mappings are retained.

1 Scope

These specifications define the descriptors for barley germplasm resources and their grading criteria.

These specifications apply to the collection, curation and conservation of barley germplasm resources; the development of data standards and data quality control specifications; and the establishment of databases and information-sharing network systems.

2 Normative references

The provisions of the following documents become provisions of these specifications through reference herein. For dated references, subsequent amendments (excluding corrections of errors) or revised editions do not apply to these specifications. However, parties entering into agreements based on these specifications are encouraged to consider whether the latest editions of these documents may be used. For undated references, the latest edition applies.

ISO 3166 Codes for the Representation of Names of Countries

GB/T 2659 Codes for the Names of Countries and Regions of the World

GB/T 2260 Codes for the Administrative Divisions of the People's Republic of China

GB/T [number not stated in the source] Codes for Institutional Affiliation

GB4404.1-1996 Seeds of Food Crops - Cereals

GB/T3543.4-1995 Rules for Agricultural Seed Testing - Germination Test

GB5519 Method for Determining Thousand-Grain Weight of Grain and Oilseeds

GB7416 Malting Barley

GB10367 Feed Barley

GB11760 Barley for Food

GB/T15796-1995 Rules for the Investigation and Forecasting of Fusarium Head Blight of Wheat and Barley

GB2905-82 Method for Determining Crude Protein in Seeds of Cereal and Leguminous Crops (Semi-micro Kjeldahl Method)

GB/T5511-1985 Inspection of Grain and Oilseeds - Method for Determining Crude Protein

GB/T5514-1985 Inspection of Grain and Oilseeds - Method for Determining Starch

GB4801-84 Method for Determining Lysine in Cereal Grains - Dye-Binding Lysine (DBL) Method

GB7649-87 Pretreatment for the Determination of Amino Acids in Cereal Grains

3 Terms and definitions

3.1 Barley

Common barley is an annual herbaceous plant of the family Gramineae and the genus Hordeum. Its scientific name is Hordeum vulgare L., and its chromosome number is 2n=2X=14.

3.2 Barley germplasm resources

Wild barley resources, landraces, bred varieties, breeding lines, genetic stocks and other materials.

3.3 Basic information

Information describing the basic attributes of barley germplasm resources, including the national unified identifier, germplasm name, scientific name, origin and germplasm type.

3.4 Morphological and biological characteristics

Characteristics of barley germplasm resources such as phenological stages, botanical morphology and yield traits.

3.5 Quality characteristics

Malting quality, brewing quality, nutritional quality and other quality characteristics of barley germplasm resources. These include grain texture, grain plumpness, hull percentage, water sensitivity, starch content, protein content, β-glucan content, extract and diastatic power.

3.6 Abiotic stress resistance

The ability of barley germplasm resources to adapt to or withstand abiotic stresses arising from soil, climate and other factors. These include cold tolerance, drought resistance, waterlogging tolerance, salt tolerance and acid tolerance.

3.7 Resistance to diseases and insect pests

The ability of barley germplasm resources to adapt to or withstand stresses caused by harmful organisms. These include barley yellow mosaic disease, barley yellow dwarf disease, Fusarium head blight, powdery mildew, leaf stripe, root rot and aphids.

4 Basic information

4.1 National unified identifier

The unique identifier for each barley germplasm accession, consisting of a prefix of three uppercase Hanyu Pinyin letters followed by a five-digit serial number.

4.2 Genebank accession number

The identifier of each barley germplasm accession in the National Long-Term Crop Germplasm Bank, consisting of “I1E” followed by a five-digit serial number.

4.3 Introduction number

The number assigned to barley germplasm when it is introduced from abroad.

4.4 Collecting number

The original number assigned to barley germplasm when it is collected in the wild.

4.5 Germplasm name

The Chinese name of the barley germplasm.

4.7 Foreign-language germplasm name

The original foreign-language name of barley germplasm introduced from abroad, or the Hanyu Pinyin name of domestic barley germplasm.

4.7 Family name

Gramineae (the grass family).

4.8 Genus name

Hordeum (the barley genus).

4.9 Scientific name

Common barley (Hordeum vulgare L.).

4.10 Country of origin

The name of the country, region or international organization to which the place of origin or collecting site of the barley germplasm belongs.

4.11 Province or first-level administrative region of origin

For domestic barley germplasm, the name of the province, municipality directly under the central government or autonomous region containing the place of origin or collecting site. For germplasm introduced from abroad, the name of the first-level administrative region in the country of origin.

4.12 Locality of origin

The names of the county, township and village of origin of domestic barley germplasm.

4.13 Elevation

The elevation of the place of origin or collecting site of the barley germplasm. Unit: m.

4.14 Longitude

The longitude of the place of origin or collecting site of the barley germplasm, expressed in degrees (°) and minutes (′), in the format DDDFF, where DDD denotes degrees and FF denotes minutes.

4.15 Latitude

The latitude of the place of origin or collecting site of the barley germplasm, expressed in degrees (°) and minutes (′), in the format DDFF, where DD denotes degrees and FF denotes minutes.

4.16 Source locality

For domestic barley germplasm, the names of the province and county from which it was obtained. For barley germplasm introduced from abroad, the name of the country, region or international organization from which it was obtained.

4.17 Original holding institution

The name of the institution that held the barley germplasm before its submission to the National Long-Term Crop Germplasm Bank.

4.18 Accession number at the original holding institution

The number assigned to the barley germplasm by the original holding institution.

4.19 Pedigree

The parents of a bred barley variety or line and the manner in which they were combined in crosses.

4.20 Breeder

The name of the institution or individual that bred the barley variety or line.

4.21 Year of development

The year in which the barley variety or line was successfully developed.

4.22 Breeding method

The breeding method used to develop the barley variety or line.

4.23 Germplasm type

Barley germplasm is classified into the following six types.

1	Wild resources
2	Landrace
3	Bred variety
4	Breeding line
5	Genetic stock
6	Other

4.24 Image

The filename of an image of the barley germplasm. File format: .jpg.

4.25 Observation site

The name of the site where the morphological and biological characteristics of the barley germplasm are observed.

5 Morphological and biological characteristics

5.1 Sowing date

The date of field sowing for the characterization and evaluation of barley germplasm resources, expressed as year, month and day in the format YYYYMMDD.

5.2 Seedling emergence date

The date after sowing when the coleoptiles of at least 50% of seeds protrude 1 cm above the soil surface, expressed as year, month and day in the format YYYYMMDD.

5.3 Heading date

The date when the spikes on at least 50% of stems have emerged 1 cm beyond the leaf sheath, excluding awns, expressed as year, month and day in the format YYYYMMDD.

5.4 Maturity date

The date when at least 80% of plants have yellowed and senesced normally and the grains have hardened so that they cannot be cut with a fingernail, expressed as year, month and day in the format YYYYMMDD.

5.5 Seedling growth habit

The growth habit of barley seedlings is classified into the following three types (Figure 1).

1	Prostrate
2	Semi-prostrate
3	Erect

Figure 1. Seedling growth habit



Prostrate Erect

5.6 Growth habit and vernalization requirement

The requirement of barley germplasm for low-temperature vernalization.

1	Winter
2	Semi-winter
3	Spring

5.7 Photoperiod response

The response of barley germplasm to day length.

1	Low sensitivity
2	Intermediate
3	Sensitive

5.8 Tillering ability

The abundance of tillers per plant in barley germplasm.

1	Strong
2	Intermediate
3	Weak

5.9 Auricle colour

The colour of the ear-like structures between the leaf blade and leaf sheath of barley germplasm.

1	White
2	Green
3	Red
4	Purple

5.10 Leaf blade length

Also termed leaf length: the distance from the tip to the sheath of the first leaf below the flag leaf. Unit: cm.

5.11 Leaf blade width

Also termed leaf width: the width at the middle of the blade of the first leaf below the flag leaf. Unit: cm.

5.12 Leaf attitude

The spatial orientation of the leaf blade after heading in barley germplasm.

1	Erect
2	Horizontal
3	Drooping

5.13 Leaf colour

The colour of the leaf blade at heading in barley germplasm.

1	Light green
2	Green
3	Dark green

5.14 Waxiness of stems and leaves

The amount of white waxy material on the stems and leaves of barley germplasm.

1	Absent or very little
2	Intermediate
3	Abundant

5.15 Plant type

The growth habit of barley plants after stem elongation and heading.

1	Compact
2	Semi-compact
3	Open

5.16 Local maturity class

The degree of earliness or lateness of barley germplasm relative to a local medium-maturing variety.

1	Very early
2	Early
3	Medium
4	Late
5	Very late

5.17 Plant height

The height of a mature plant from the tillering node of the main stem to the tip of the spike, excluding awns. Unit: cm.

5.18 Stem diameter

The diameter at the middle of the second aboveground internode of the main stem. Unit: mm.

5.19 Spikes per plant

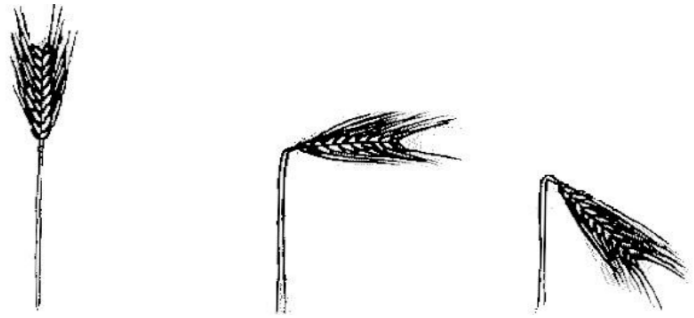
The average number of spikes per plant in barley germplasm. Unit: spikes.

5.20 Spike attitude at maturity

The orientation of the mature spike on the stem (Figure 2).

1	Erect
2	Horizontal
3	Drooping

Figure 2. Spike attitude at maturity



5.21 Spike and awn colour

The colour of the spike and awns at maturity in barley germplasm.

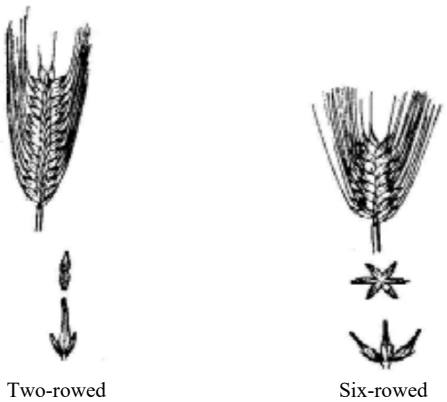
1	Yellow
2	Grey
3	Purple
4	Brown
5	Black

5.22 Row type

Barley germplasm is classified into three row types according to the fertility of the spikelet triplets on the rachis (Figure 3).

1	Two-rowed
2	Intermediate
3	Six-rowed

Figure 3. Row type



5.23 Spike length

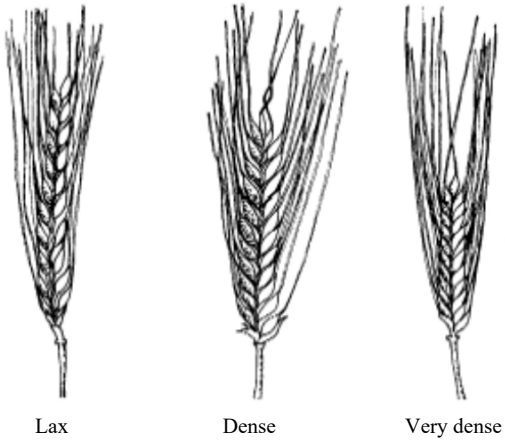
The distance from the base of the rachis to the tip of the spike, excluding awns. Unit: cm.

5.24 Spike density

The density of spikelet triplets on the rachis in barley germplasm. Three classes are distinguished according to the number of spikelet-bearing rachis segments within a 4 cm length in the middle of the spike (Figure 4).

1	Lax
2	Dense
3	Very dense

Figure 4. Spike density



5.25 Spike branching

Whether secondary spikelets develop on the rachis or on the spikelet triplets of barley germplasm.

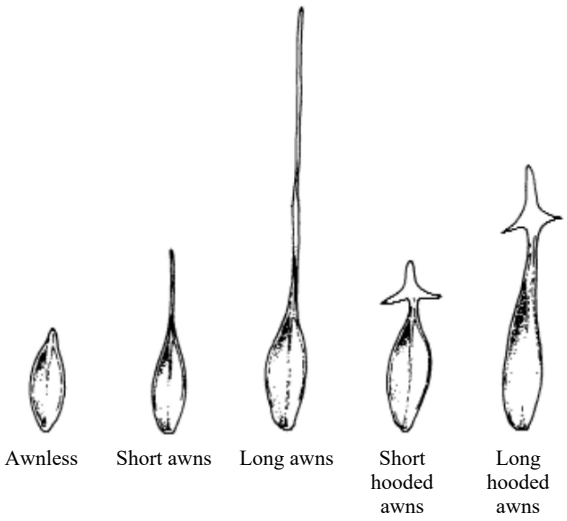
1	Unbranched
2	Branched

5.26 Awn type

The length and shape of the awns on the central and lateral spikelets of the spike. Sixteen types are distinguished (Figure 5).

0	Awnless
1	Rudimentary awns
2	Awns as long as the spike
3	Short awns
4	Long awns
5	Long central awns; lateral spikelets awnless
6	Long central awns; rudimentary lateral awns
7	Long central awns; short lateral awns
8	Short central awns; lateral spikelets awnless
9	Rudimentary central awns; lateral spikelets awnless
10	Sessile hoods (no awn neck)
11	Short hooded awns
12	Long hooded awns
13	Long hooded central awns; short hooded lateral awns
14	Hooded central awns; rudimentary lateral awns
15	Hooded central awns; lateral spikelets awnless

Figure 5. Awn type



5.27 Awn roughness

The presence or absence of tooth-like projections on the awns of barley germplasm (Figure 6).

1	Rough, serrated awns
2	Smooth awns

Figure 6. Awn roughness



Smooth awns Rough, serrated awns

5.28 Lateral spikelets

The degree of development or reduction of the lateral spikelets in two-rowed barley germplasm.

1	Lateral spikelets present
2	Lateral spikelets absent

5.29 Glume width

The width at the middle of the glume in barley germplasm.

1	Narrow
2	Wide

5.30 Rachilla hair length

The length of hairs on the rachilla after threshing (Figure 7).

1	Short
2	Long

Figure 7. Rachilla hair length



Short Long

5.31 Grains per spike

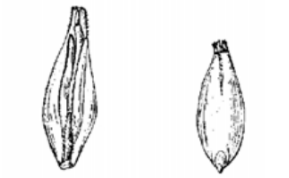
The actual number of grains set on each spike. Unit: grains.

5.32 Hull adherence

Whether the grain separates naturally from the hull at maturity (Figure 8).

1	Hulled
2	Hull-less

Figure 8. Hull adherence



Hulled barley Hull-less barley

5.33 Grain colour

The colour of mature barley grains.

1	Yellow
2	Blue
3	Purple
4	Brown
5	Black

5.34 Grain shape

The shape of mature barley grains is classified into four types (Figure 9).

1	Oblong
2	Ovate
3	Elliptic
4	Round

Figure 9. Grain shape



Oblong Ovate Elliptic Round

5.35 Thousand-grain weight

The weight of 1,000 naturally air-dried barley grains. Unit: g.

5.36 Seed dormancy period

The time from maturity until barley seeds can germinate normally. Unit: d.

5.37 Germination percentage

The proportion of a given number of barley seeds that germinate under suitable temperature and moisture conditions. Expressed as a percentage (%).

6 Quality characteristics

6.1 Grain plumpness

The percentage of barley grains retained on a 2.5 mm slotted sieve. Expressed as a percentage (%).

6.2 Hull percentage

Hull weight as a proportion of the total grain weight of hulled barley. Expressed as a percentage (%).

6.3 Grain texture

Grain texture is classified according to the proportion of vitreous tissue in the grain endosperm.

1	Hard
2	Semi-hard
3	Soft

6.4 Starch content

The proportion of crude starch in a unit weight of barley grain. Expressed as a percentage (%).

6.5 Protein content

The proportion of protein in a unit weight of barley grain. Expressed as a percentage (%).

6.6 Lysine content

The proportion of lysine in a unit weight of barley grain. Expressed as a percentage (%).

6.7 Extract

The amount of water-soluble substances produced by mashing a unit weight of malting-barley flour of a specified fineness under specified temperature conditions. Expressed as a percentage (%).

6.8 β-Glucan content

The proportion of β-glucan in a unit weight of barley grain. Expressed as a percentage (%).

6.9 Diastatic power

The capacity of 100 g of anhydrous malt from malting barley to hydrolyse soluble starch to maltose within 30 min at 20 °C. Unit: W.K (production of 1 g of maltose equals 1 W.K).

6.10 Water sensitivity

The physiological inhibition of germination caused by excess water during the germination of barley seeds.

1	None
3	Very slight
5	Slight
7	Moderate
9	Severe

7 Abiotic stress resistance

7.1 Cold tolerance

The ability of barley germplasm to tolerate or resist injury caused by low temperatures.

1	Highly tolerant
3	Tolerant
5	Moderately tolerant
7	Intolerant
9	Highly intolerant

7.2 Drought resistance

The ability of barley germplasm to tolerate or resist soil drought.

1	Highly tolerant
3	Tolerant
5	Moderately tolerant
7	Intolerant
9	Highly intolerant

7.3 Waterlogging tolerance

The ability of barley germplasm to tolerate or resist waterlogging injury.

1	Highly tolerant
3	Tolerant
5	Moderately tolerant
7	Intolerant
9	Highly intolerant

7.4 Salt tolerance

The ability of barley germplasm to tolerate or resist excessive salt levels in the soil.

1	Highly tolerant
3	Tolerant
5	Moderately tolerant
7	Intolerant
9	Highly intolerant

7.5 Acid tolerance

The ability of barley germplasm to tolerate or resist injury caused by soil acidity.

1	Highly tolerant
3	Tolerant
5	Moderately tolerant
7	Intolerant
9	Highly intolerant

8 Resistance to diseases and insect pests

8.1 Resistance to barley yellow mosaic disease

Barley yellow mosaic disease is a major disease caused by Barley yellow mosaic virus (BaYMV) and Barley mild mosaic virus (BaMMV), transmitted by the soil-borne vector Polymyxa graminis. Resistance of barley germplasm to this disease is graded as follows.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.2 Resistance to barley yellow dwarf disease

Barley yellow dwarf disease is a major disease of barley caused by Barley yellow dwarf virus (BYDV). Resistance of barley germplasm to this disease is graded as follows.

1	Resistant (R)
3	Tolerant (T)
5	Moderately tolerant (MT)
7	Susceptible (S)
9	Highly susceptible (HS)

8.3 Resistance to Fusarium head blight

Fusarium head blight is a major disease of barley caused by Fusarium graminearum. Resistance of barley germplasm to this disease is graded as follows.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.4 Resistance to powdery mildew

Powdery mildew is a major disease of barley caused by Erysiphe graminis f. sp. hordei. Resistance of barley germplasm to this disease is graded as follows.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.5 Resistance to leaf stripe

Barley leaf stripe is a major disease of barley caused by Pyrenophora graminea. Resistance of barley germplasm to this disease is described as follows.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.6 Resistance to root rot

Barley root rot is caused by the pathogens Helminthosporium sativum and Fusarium spp. Resistance of barley germplasm is graded as follows.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.7 Resistance to net blotch

Barley net blotch is caused by the ascomycete Pyrenophora teres and mainly affects the leaves. Resistance of barley germplasm is graded as follows.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.8 Resistance to aphids

The principal aphids damaging barley in China are Toxoptera graminum (Rondani), Macrosiphum granarium (Kirby) and Rhopalosiphum padi (Linnaeus). Resistance of barley germplasm to aphids is graded as follows.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

9 Other characteristics

9.1 Principal uses

The principal uses of barley grain.

1	Food
2	Feed
3	Brewing

9.2 Mutants

Barley mutations affecting special traits, arising spontaneously or through induced mutagenesis.

9.3 Near-isogenic lines

Special barley genetic materials differing only at a particular gene.

9.4 Morphological markers

Genes for marker traits named according to the international barley gene nomenclature.

9.5 Chromosomal variation

Variation in chromosome number, morphology and structural characteristics in barley germplasm.

9.6 Biochemical markers

Protein and enzyme markers of barley germplasm.

9.7 Molecular markers

DNA molecular markers for major traits of barley germplasm, including restriction fragment length polymorphism (RFLP), random amplified polymorphic DNA (RAPD), amplified fragment length polymorphism (AFLP), simple sequence repeat polymorphism (SSR) and single nucleotide polymorphism (SNP).

9.8 Remarks

Detailed explanations of special barley descriptors or codes.

Translator's notes on the source

2: The source omits the number of the GB/T reference on institutional affiliation. The title of GB11760 is translated descriptively as Barley for Food; no missing standard number or edition has been supplied.

4.7: The Chinese source assigns 4.7 to both Foreign-language germplasm name and Family name and contains no 4.6. This numbering is retained.

4.14-4.15: Coordinates are expressed in degrees and minutes. The source formats DDDFF for longitude and DDFF for latitude are retained; the source does not specify hemisphere signs.

5.24: The source gives no numerical boundaries between the three spike-density classes.

5.26; Figure 5: Hooded terminology is supported by comparison with the original IPGRI (1994) lemma awn/hood illustrations. The source retains separate neckless, short and long categories, but does not illustrate code 10 or define numerical length limits. Sessile hoods (no awn neck) is a morphological rendering of code 10; the Chinese categories must not be replaced by the different IPGRI coding scheme.

5.30; Figure 7: The Chinese text uses the term for rachis, but its Figure 7 reproduces the rachilla-hair illustration in IPGRI (1994), section 7.2.12 and Figure 9. Rachilla is therefore used here to identify the illustrated organ accurately. The source codes and the observation after threshing are retained.

6.7: The source specifies malting-barley flour rather than malt flour; the material and procedure have not been silently changed.

6.9: The source definition of W.K, including 20 °C, 100 g, 30 min and 1 g maltose per W.K, is retained without substitution of a different assay definition.

8.1; 8.4; 8.6; 8.8: Historical taxa are retained, with verified orthographic and author corrections: Erysiphe graminis f. sp. hordei (EPPO); Toxoptera graminum (Rondani), Macrosiphum granarium (Kirby) and Rhopalosiphum padi (Linnaeus) (GBIF). Polymyxa receives conventional genus capitalization. In 8.6, and separates two pathogen names printed without punctuation. The aphids are identified by their scientific names.